a Output agreement in each workflow, model and replicate



b Workflow summaries

IWC workflow	Galaxy 5.5 1	Galaxy 5.5 2	Galaxy 5.5 3	Galaxy Sol 1	Galaxy Sol 2	Galaxy Sol 3	Galaxy Luna 1	Galaxy Luna 2	Galaxy Luna 3	Galaxy DeepSeek 1	Galaxy DeepSeek 2	Galaxy DeepSeek 3	Custom code 5.5 1	Custom code 5.5 2	Custom code 5.5 3	Custom code Sol 1	Custom code Sol 2	Custom code Sol 3	Custom code Luna 1	Custom code Luna 2	Custom code Luna 3	Custom code DeepSeek 1	Custom code DeepSeek 2	Custom code DeepSeek 3	Mean % (runs <99%) Galaxy	Mean % (runs <99%) Custom code
001 Read QC and trimming	100	100	100	100	100	100	99 to <100	99 to <100	99 to <100	99 to <100	99 to <100	99 to <100	100	100	100	100	100	100	100	100	100	100	100	99 to <100	**99.89 (0)**	**99.97 (0)**
002 RNA-seq differential expression	100	100	99 to <100	100	100	100	99 to <100	100	100	100	100	100	99 to <100	100	99 to <100	100	100	100	100	100	100	100	99 to <100	99 to <100	**99.95 (0)**	**100.00 (0)**
003 Host-read filtering	100	100	100	100	100	100	99 to <100	100	100	99 to <100	100	100	100	99 to <100	100	100	100	100	100	100	100	100	100	100	**100.00 (0)**	**99.96 (0)**
005 Amplicon denoising	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	100	100	100	0	99 to <100	0	90 to <99	90 to <99	90 to <99	99 to <100	100	99 to <100	99 to <100	100	100	**95.86 (9)**	**79.67 (9)**
006 ATAC-seq accessibility	100	100	99 to <100	99 to <100	100	99 to <100	100	100	99 to <100	99 to <100	100	100	100	99 to <100	99 to <100	90 to <99	99 to <100	100	100	100	99 to <100	100	99 to <100	100	**99.78 (0)**	**99.03 (1)**
007 Mitogenome assembly	90 to <99	90 to <99	100	100	100	100	99 to <100	100	99 to <100	100	99 to <100	100	90 to <99	90 to <99	0	90 to <99	90 to <99	99 to <100	90 to <99	90 to <99	90 to <99	99 to <100	99 to <100	99 to <100	**98.97 (5)**	**90.70 (10)**
008 AMR gene detection	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	**100.00 (0)**	**100.00 (0)**
009 Peptide verification	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	90 to <99	0	0	100	100	100	90 to <99	0	0	100	90 to <99	90 to <99	90 to <99	90 to <99	0	99 to <100	90 to <99	100	**95.64 (10)**	**93.04 (10)**
010 Pseudobulk differential expression	100	90 to <99	100	100	100	90 to <99	100	100	90 to <99	100	100	90 to <99	100	90 to <99	90 to <99	100	100	90 to <99	100	100	100	100	100	0	**99.79 (1)**	**91.02 (3)**
011 BioProject data retrieval	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	**100.00 (0)**	**100.00 (0)**

Columns 1–3 are replicates. Models are GPT-5.5, GPT-5.6 Sol, GPT-5.6 Luna and DeepSeek V4 Pro.

Each workflow mean uses 12 runs per policy. A cross marks zero agreement.